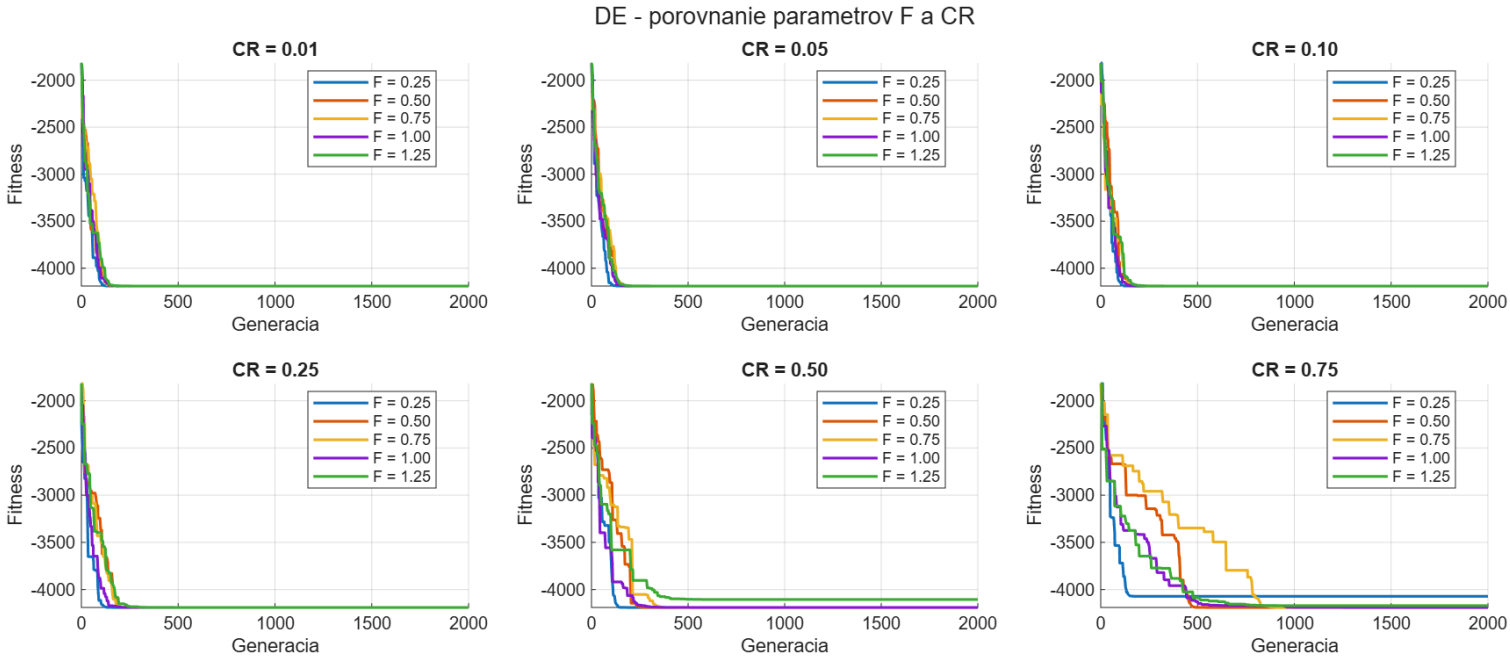
Slovenská Technická Univerzita – Fakulta elektrotechniky a informatiky, Ilkovičova 3, 812 19 Bratislava

Diferenciálna evolúcia

Umelá inteligencia 2, zadanie č. 5

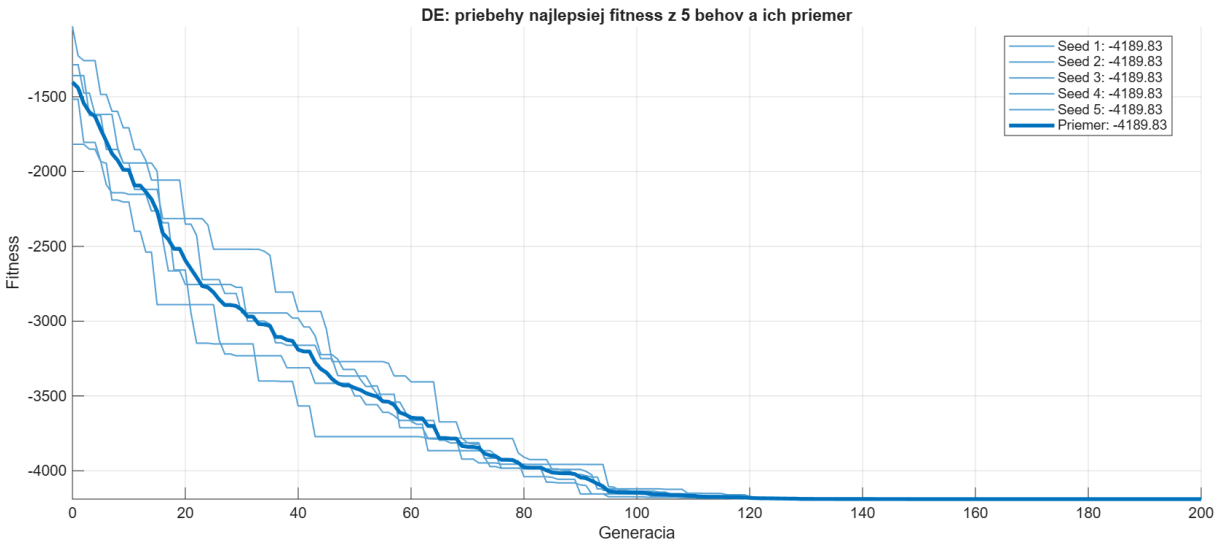
Autor: **Matúš Nebus**
Dátum vypracovania: **20.10.2026**
Ročník štúdia: **1. Ing.**
Študijný program: **Robotika a kybernetika**

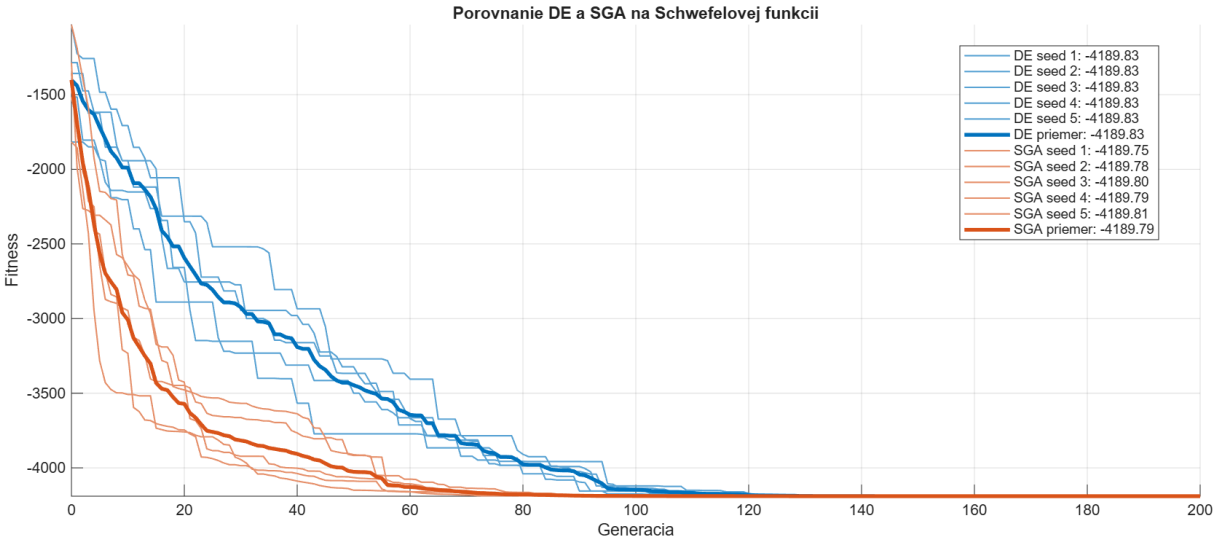
Experimentálne hľadanie hodnôt parametrov F a CR:



Za najlepšie parametre DE som určil: CR = 0.05, F = 0.25.

```
%nastavenia SGA
pmut_global = 0.3;
pmut_local = 0.3;
Amp_local = 5*ones(1,nvars);
bestNumsSGA = ones(1,2);
RandOldCountSGA = 10;
work1CountSGA = 18;
work2CountSGA = 20;
popSize = 50;
ngen = 200;
```





Beh	DE	SGA
"Seed 1"	-4189.8	-4189.7
"Seed 2"	-4189.8	-4189.8
"Seed 3"	-4189.8	-4189.8
"Seed 4"	-4189.8	-4189.8
"Seed 5"	-4189.8	-4189.8
"Priemer"	-4189.8	-4189.8
"Smerodajna odchylka"	0.00028827	0.023101

```
==========================================
NAJLEPSIE NAJDENE VEKTORY
==========================================
```

Algoritmus	Seed	Fitness	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
"DE"	4	-4189.8	420.96	420.97	420.98	420.96	420.97	420.97	420.97	420.96	420.97	420.97
"SGA"	5	-4189.8	420.89	421	421.02	420.84	420.6	421.03	420.97	421.06	421	421.02